NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END SEMESTER EXAMINATION, JULY – DEC 2024

BTech: 1st Semester (Section I, II, III)
Course Name: Information Technology Workshop **Course Code: CS14106**
Maximum Time: 3 hours **Maximum Marks: 60**

Instructions:

1. Attempt all questions & sub questions must be answered sequentially at one place.
2. Assume any suitable data, if necessary.
3. The marks, Course Outcome (CO) and Bloom's Level (BL) related to questions are mentioned on the right-hand side margin.

		Question	Marks	CO	BL
1	a.	Name and describe in brief the basic elements that make up a computer system.	5+5	CO1	1, 2
	b.	Explain the role of a Loader in a computer system. How does the loader load an executable program into memory, and what is the significance of address binding in this process?			
2	a.	Explain in brief the function of the Control Unit (CU). How does the CU decode the instruction from memory and send control signals to direct the ALU, registers, and other components?	5+5	CO2, 3	2
	b.	Explain the characteristics and limitations of Dynamic RAM (DRAM) and Static RAM (SRAM). How do these two types of RAM differ in terms of speed, volatility, and power consumption?			
3	a.	Define CPU registers, how do registers contribute to the overall performance and speed of the processor compared to other types of memory?	5+3+2	CO3, 4	1, 6
	b.	Give the steps to Convert Text to a Table in Microsoft Word.			
	c.	Differentiate between a formula and a function as used in Microsoft Excel.			
4	a.	How do the various data transmission modes—namely simplex, half-duplex, and full-duplex—differ in terms of their communication flow, speed, bandwidth utilization, and practical applications, and what are the advantages and limitations of each mode in different types of network environments.	5+5	CO5	1, 2
	b.	Define LAN, MAN, and WAN, and describe the key differences in their scope, geographical coverage, data transmission rates, and typical applications.			
5	a.	Explain in brief Ring Topology, Bus Topology, and Star Topology, including how data is transmitted, the role of devices in the topology, and the setup and layout of each.	5+5	CO5	2
	b.	Explain in brief the different types of VPN. What are the benefits of a VPN connection?			
6	a.	How do various types of malicious software, such as viruses, worms, ransomware, spyware, and Trojans, differ in their mechanisms of infection, spread, and damage?	5+5	CO6	2
	b.	Explain in brief Hacking and Firewall.			

Wish you all the best